KEROLOS YOUSEF ZAHY

DevOps Engineer • DevSecOps • Cloud Infrastructure (AWS) • Kubernetes • GitOps • CI/CD

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• github.com/Kerolos129Yousef

PROFESSIONAL SUMMARY

DevOps / DevSecOps Engineer with hands-on delivery of secure CI/CD pipelines, Kubernetes GitOps deployments, and cloud infrastructure automation on AWS. Built production-grade systems including a 7-stage Gitea Actions pipeline with SBOM generation (Syft/SPDX-JSON), ArgoCD self-healing GitOps, and Argo Rollouts canary deployments as part of graduate coursework at Queen's University (MSc, Computer Engineering). Proficient in Docker, Kubernetes, Terraform, Ansible, Prometheus, and Grafana. Seeking a DevOps or DevSecOps role where strong fundamentals and a bias toward automation create real infrastructure value. Additionally familiar with MLOps practices including model fine-tuning, distributed data pipelines, and LLM deployment on AWS.

EXPERIENCE

Cloud / DevOps Engineer — Intern | Digilians — Cairo, Egypt ***Sep 2025 – Jul 2026***

- Designed and implemented end-to-end CI/CD pipelines using GitHub Actions and Gitea Actions — automating Docker image builds, pushes to Docker Hub, and zero-touch SSH deployments to AWS EC2, eliminating all manual release steps from commit to production.
- Provisioned and managed cloud infrastructure on AWS (EC2, VPC, subnets, security groups) using Terraform (IaC) and automated server configuration with Ansible playbooks and Ansible Vault for secure secrets management.
- Containerized multi-service applications (Node.js, React, FastAPI, MongoDB) using Docker and Docker Compose with multi-stage hardened builds, reducing production image size by ~40–50% and improving deployment consistency.
- Deployed and managed Kubernetes workloads (k3d) with ArgoCD GitOps — authoring manifests, Kustomize overlays, and Argo Rollouts canary strategies; configured ArgoCD self-heal sync to automatically reconcile config drift within 30 s of a Git push, enabling zero-downtime releases.
- Configured Nginx as a reverse proxy with security headers, WebSocket support, and immutable SHA-tagged deployments, enabling rollbacks in under 2 minutes.
- Instrumented applications with Prometheus custom metrics (counters, histograms) and provisioned Grafana dashboards for real-time observability of request rates, error rates, and p50/p95/p99 latency.
- Introduced SBOM generation (Syft/SPDX-JSON) and CI security gates (ruff lint, mypy type-checking, pytest coverage >70%) into team pipelines, establishing a baseline supply-chain security posture.

Earlier Experience

- **Android Developer Intern, Information Technology Institute / ITI** (Jul–Sep 2023): Developed Android mobile applications using Java, focusing on mobile application architecture and UI/UX principles; applied Agile development practices.
- **Frontend Developer Intern, Route Company** (Mar–Jul 2022): Built and styled multiple real-world web applications using React, JavaScript, Bootstrap 5, and REST APIs; delivered a full-featured Product Management System with CRUD operations and a responsive dashboard.

KEY PROJECTS

Secure CI/CD + GitOps + Kubernetes — Sentiment API ***May 2026***

Gitea Actions · k3d · Kustomize · ArgoCD · Argo Rollouts · Prometheus · Grafana · FastAPI | Queen's University — CISC-814 | github.com/Kerolos129Yousef/ml-sentiment-app

- **End-to-End CI pipeline:** Built a 7-stage Gitea Actions workflow (ruff lint, mypy type-checking, pytest with >70% coverage, multi-stage Docker build, image push, SBOM generation via Syft/SPDX-JSON) — automating the full build-test-push-deploy cycle and eliminating all manual release steps from commit to deployment; multi-stage builds cut image size ~40%.
- **Kubernetes & GitOps:** Authored 6 Kubernetes manifests (Deployment, Service, ConfigMap, Rollout, canary Service, ServiceMonitor) with liveness/readiness probes, resource limits, and Kustomize overlays; wired to ArgoCD with automated self-heal sync — configured to detect and reconcile config drift within 30 s of a Git push across all manifests.
- **Canary deployment:** Replaced standard Deployment with an Argo Rollout executing a 5-step canary strategy (20 > 40 > 60 > 80 > 100%) across ~4 replica pods, enabling zero-downtime releases with automatic rollback capability.
- **Observability:** Instrumented FastAPI app with 3 Prometheus metrics (request counter, label counter, latency histogram); provisioned a Grafana dashboard (4 panels: request rate, 5xx error rate, p50/p95/p99 latency, pod CPU/memory) — providing real-time visibility into >95% of prediction traffic.
- **Infrastructure as Code:** Entire stack (cluster, ArgoCD, Argo Rollouts, kube-prometheus-stack) reproducible from a single manage-cluster.sh create script with zero manual kubectl intervention post-setup.

AI Study Companion — DevOps & Cloud Infrastructure Engineer ***2026***

GitHub Actions · Docker · Terraform · Ansible · AWS EC2 · Nginx · Ansible Vault | github.com/Kerolos129Yousef/AI-Study-Companion

- Designed and implemented an end-to-end CI/CD pipeline using GitHub Actions, Docker Hub, and AWS EC2, enabling automated production releases in under 5 minutes.
- Provisioned AWS infrastructure using Terraform and automated server configuration through Ansible, reducing environment setup time from 1–2 hours to less than 10 minutes.
- Containerized frontend and backend services using Docker and Docker Compose with multi-stage builds, improving deployment consistency and reducing production image size by approximately 50%.
- Implemented secure secrets management using Ansible Vault and GitHub Secrets to manage 10+ application secrets and environment variables without exposing credentials in source control.

- Configured Nginx as a reverse proxy with security headers, WebSocket support, and immutable SHA-tagged container deployments, enabling rollbacks in under 2 minutes.

End-to-End Cloud-Based Chat Assistant ***2026***

AWS (EC2, EMR, VPC) · Terraform · Apache Spark · PyTorch · QLoRA · llama.cpp · Streamlit | github.com/Kerolos129Yousef/Tech_chat_bot

- **Infrastructure as Code:** Architected and deployed a fully isolated AWS network topology using Terraform, configuring a custom VPC, private/public subnets, security groups, and an Internet Gateway.
- **Distributed Big Data:** Provisioned an AWS EMR cluster to run distributed Apache Spark (PySpark) jobs, preprocessing and cleaning over 150,000 StackExchange/RedPajama QA records.
- **PEFT / QLoRA:** Fine-tuned Meta-Llama-3-8B via QLoRA (4-bit quantization) in an isolated Docker container using an NVIDIA RTX 2000 Ada GPU.
- **Production Deployment:** Converted fine-tuned weights to GGUF format and deployed on AWS EC2 (m5.xlarge, Ubuntu 22.04 LTS) via a CPU-optimised llama.cpp inference container.
- **Full-Stack Interface:** Developed and hosted an interactive Streamlit web application connected to the backend inference API.

EDUCATION

MSc, Computer Engineering | Queen's University — Kingston, Canada ***Sep 2025 – Jul 2026***

Focus: Distributed Systems, Cloud-Native Architectures, Software Engineering. Completed CISC-814 (graduate-level software engineering).

BSc, Computer Science & Artificial Intelligence — Excellent with Honors (GPA 3.55) | Helwan University — Cairo, Egypt ***Sep 2019 – Jun 2023***

Specialization: Medical Informatics. Capstone: Non-Invasive Anemia Detection (deep learning, mobile).

Training & Competitions: ECPC — Egyptian Collegiate Programming Contest, national-level participant (Aug 2021) • IEEE Helwan — C programming & programming in medical technology (Feb–Mar 2020)

TECHNICAL SKILLS

Cloud Platforms	AWS (EC2, S3, VPC, Subnets, Security Groups, EMR, IAM, ECR)
Containers	Docker, Docker Compose, Docker Hub, Kubernetes (k3d, manifests, liveness/readiness probes, resource limits, RBAC)
GitOps & CD	ArgoCD, Argo Rollouts (canary / blue-green), Kustomize, Helm
CI/CD	GitHub Actions, Gitea Actions, Git (branching strategies, PR workflows)
IaC & Config	Terraform (VPC, EC2, EMR provisioning), Ansible (playbooks, roles, Ansible Vault)
DevSecOps	Syft (SBOM / SPDX-JSON), Trivy (CVE scanning), Docker Scout, multi-stage hardened Docker builds, CI security gates (lint, type-check, coverage enforcement), supply-chain security
Observability	Prometheus (custom metrics: counters, histograms), Grafana (dashboards, alerts)
Programming	Python, Bash / Shell, JavaScript, FastAPI
Databases	MongoDB, PostgreSQL, SQL
Methodologies	Linux administration, GitOps, Agile / Scrum, DevSecOps principles

CERTIFICATIONS

Cloud & Infrastructure

- AWS Certified Cloud Practitioner — Amazon Web Services (2026)

AI & Machine Learning

- Fundamentals of Deep Learning — NVIDIA (Mar 2026)
- Supervised Machine Learning: Regression & Classification — DeepLearning.AI (Feb 2026)

DevOps & Linux

- Docker for Beginners with Hands-on Labs — KodeKloud (Dec 2025)
- Docker Basics for DevOps — KodeKloud (Nov 2025)
- Git Basics for DevOps — KodeKloud (Nov 2025)
- DevOps Prerequisite Course — KodeKloud (Nov 2025)

SOFT SKILLS

Problem-Solving Under Pressure • Cross-Functional Collaboration • Attention to Detail • Continuous Learning & Adaptability • Ownership & Accountability • Time Management • Documentation & Knowledge Sharing